



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations
 May 2024

RI2109-1 – MICROCONTROLLER AND ITS APPLICATIONS
 (RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- What is the primary purpose of a microcontroller?
 A) Data storage B) General-purpose computing
 C) Signal processing D) Control of embedded systems
- How many bits are in the data bus of an 8-bit microcontroller?
 A) 4 B) 16
 C) 8 D) 32
- In 8051 architecture, where is the stack pointer located?
 A) RAM B) ROM
 C) Special Function Register D) Data Bus
- Which addressing mode is commonly used for accessing constants in 8051?
 A) Immediate addressing mode B) Register addressing mode
 C) Indirect addressing mode D) Direct addressing mode
- In serial communication, what is the purpose of I2C and SPI protocols?
 A) Data encryption B) Error detection
 C) Communication between multiple devices D) Power management
- Which module is responsible for serial communication in the 8051 microcontroller?
 A) Timer/Counter B) Interrupts
 C) UART D) I2C
- What is the primary purpose of timer interrupts in a microcontroller?
 A) Counting external events B) Handling keyboard inputs
 C) Managing power consumption D) Controlling display units
- In 8051 microcontroller, which modes are available for Timers/Counters?
 A) Mode 1 and Mode 2 B) Mode A and Mode B
 C) Mode X and Mode Y D) Mode Alpha and Mode Beta
- In 8051 architecture, where is the program counter (PC) used?
 A) ALU B) Stack
 C) I/O ports D) Memory
- Which register is used for addressing external data memory in 8051 programming?
 A) ACC B) B
 C) DPTR D) PSW
- What is the role of the instruction CJNE A, #10, LABEL in 8051 programming?
 A) Compare A with 10, and jump to LABEL if not equal B) Compare A with 10, and jump to LABEL if equal
 C) Copy the value 10 to register A D) Jump to LABEL unconditionally
- What is the function of the instruction MOV A, @R0 in 8051 programming?
 A) Move the value at the address in R0 to register A B) Move the value in register R0 to register A
 C) Move the value at the address pointed by R0 to register A D) Move the value at the address in register A to register R0

13. What is the baud rate in serial communication?
 - A) The number of bits per character
 - B) The number of characters per second
 - C) The time taken to transmit one bit
 - D) The time taken to transmit one character
14. How is the Serial Data Output (TXD) pin configured in 8051 serial communication?
 - A) Input
 - B) Output
 - C) Bidirectional
 - D) Tri-State
15. In 8051 serial communication, which register is used to load the serial data to be transmitted?
 - A) SBUF
 - B) SCON
 - C) TBUF
 - D) TXD
16. What is the main characteristic of ARM architecture?
 - A) 8-bit data bus
 - B) 16-bit data bus
 - C) 32-bit data bus
 - D) 64-bit data bus
17. In ARM processor modes, which mode allows access to all registers?
 - A) User mode
 - B) Supervisor mode
 - C) Interrupt mode
 - D) System mode
18. How many processor modes are there in ARM architecture?
 - A) 2
 - B) 4
 - C) 3
 - D) 5
19. What is the primary role of the data processing instructions in ARM?
 - A) Transfer data between registers
 - B) Control the program flow
 - C) Manage I/O ports
 - D) Perform arithmetic and logic operations
20. How many general-purpose registers are available in ARM architecture?
 - A) 16
 - B) 64
 - C) 32
 - D) 8

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) With a neat architectural diagram explain the functions of each block in 8051 microcontrollers.	8	L*2	1	1
	b) Explain Logical instructions in 8051 microcontrollers with an example for each case.	8	L3	2	1
2.	a) With the neat diagram, explain 128 bytes RAM memory organization of 8051 microcontroller with address in detail.	8	L2	1	1
	b) Write a program to find the sum of all the numbers. The result must be in BCD. Assume that 5 BCD data items are stored in RAM locations starting at 40H.	8	L3	2	1
3.	a) Write a note on: Stack, Stack Pointer, Program counter, DPTR with examples.	8	L2	1	1
	b) For 8051 system of 11.0592MHz, find out how long it takes to execute each Instructions. i) MOV R3, #55 ii) DEC R3 iii) DJNZ R2, TARGET iv) MUL AB v) NOP	8	L3	2	1
Unit – II					
4.	a) Write a microcontroller 8051 program to count the number of 0s and 1s in a byte and store the results in R3 and R4 respectively also explain the program using flowchart.	8	L3	3	1
	b) Draw a neat block diagram of Timer 1 in Mode1 and Mode2 of 8051 microcontrollers. Write the steps to program timers in Mode 1.	8	L2	4	1

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|-------|---|---|----|---|---|
| 5. a) | Explain the programming I/O ports in 8051 microcontrollers with a neat diagram and explain the operation of I/O port 0. | 8 | L2 | 3 | 1 |
| b) | Assume XTAL = 22 MHz, write a program to generate a triangular waveform on pin P1.5. use Timer1 Mode1. | 8 | L3 | 4 | 1 |
| 6. a) | Interface 2-digit common cathode seven segment display to microcontroller 8051 and display '123' using suitable ports and explain the same. | 8 | L3 | 3 | 1 |
| b) | Explain I2C and SPI communication protocols also compare and contrast both. | 8 | L2 | 4 | 1 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | Draw a neat architecture diagram to represent 3 stage pipelines in ARM processor. Explain single cycle instruction with relevant figures. | 8 | L2 | 5 | 1 |
| b) | Explain any 5 Branch instruction of ARM with examples. | 8 | L2 | 5 | 1 |
| 8. a) | Explain the function of following instructions with an example. | | | | |
| | i) MOV | | | | |
| | ii) MVN | | | | |
| | iii) MLA | | | | |
| | iv) BIC | | | | |
| | v) RSB | 8 | L3 | 5 | 1 |
| b) | Explain the memory organization of ARM in detail with figure. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
